Fix Log — Mobile, Readability & Performance Pass

Two live properties audited:

- **Portfolio (case studies):** https://yuguda999.github.io/ys-dev/
- **Personal site:** https://yuguda.netlify.app/

Audit method: rendered both at 360px / 390px / tablet / desktop widths in a headless browser, computed WCAG contrast ratios for every colour pair in the CSS rather than eyeballing them, requested every link on every page and recorded the status code, and measured real transferred bytes per asset. Then opened both on an actual phone (browser emulation is not the same thing — see "checked on a real phone" below).

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1. Nav unusable on a phone

Before: the header was `padding: 20px 40px` with nav links spaced by `margin-left: 24px`. At 360px the brand wrapped onto its own line and the four links crowded the next one. Worse, the links were bare text with no padding — roughly 20px tall, well under the ~44px minimum for a reliable thumb tap. Easy to hit the wrong case study.

After: header padding reduced to `14px 24px` (`10px 16px` under 600px), and each link now carries `padding: 10px` instead of a left margin, so the clickable box itself is tall enough to tap. Links wrap as a flex row. Same treatment applied to the "← All case studies" back link and the footer links, which had the same problem.

2. Headings ate the screen

Before: `.hero h1 { font-size: 2.2em }` — a fixed size regardless of viewport. "Yuguda Muhammed Shamsudeen" ran to three lines on a phone before any content was visible.

After: `font-size: clamp(1.75rem, 6vw, 2.4rem)`. The heading scales with viewport width but is floored at 1.75rem and capped at 2.4rem, so it stays two lines on mobile and unchanged on desktop. Case-study headings got the same treatment.

3. Work captures illegible on mobile

Before: the CampusPQ API screenshot was 1905px wide and the vLLM log capture 1502px, both rendered with `width: 100%` into a ~340px column. The terminal log text and API endpoint names became unreadable mush — the exact evidence those case studies rest on.

After: each screenshot is wrapped in a link to the full-size image with a "Tap to open the full-size capture" caption, so a phone reader is one tap from a legible version instead of squinting. Also added explicit `width` / `height` attributes so the browser reserves the right space before the image loads (no layout jump), and `loading="lazy"` so below-the-fold images don't block first paint.

4. Oversized images

Before: 741 KB across three PNGs, the worst being the vLLM log at 498 KB.

After: downscaled to a 1400px maximum and colour-quantized (screenshots have small palettes, so this is nearly lossless for them) — **741 KB → 348 KB, 53% lighter**. Checked the result at full size: terminal text is still sharp.

5. The run capture was 52 MB

Before: the capstone demo video was a 1080p60 screen recording at 6.1 Mbps. Nothing loads it on page view (`preload="metadata"`), but anyone tapping play on mobile data pulled 52 MB.

After: re-encoded to 30fps at CRF 26 with `-movflags +faststart` — **52 MB → 1.9 MB, 96% smaller**. 60fps is pointless for a terminal recording, and `faststart` moves the file index to the front so playback can begin before the download finishes. Content is untouched (still the same raw, unedited run); I compared frames before and after to confirm the terminal text is still readable.

6. Contrast failure

Before: captions and footer text used `#6B6B80` on the `#08080C` background — **3.85:1**, under the 4.5:1 WCAG AA minimum for body text.

After: `#8A8A9E`, now **5.91:1**. Every other pair on both sites was computed and passes: body text 17:1 and 16.98:1, muted text 4.55–6.72:1, links 4.94–11.06:1.

7. The whole git history was public (worst find)

Before: the personal site is deployed by dragging its folder onto Netlify Drop — which uploaded the `.git` directory along with it. `https://yuguda.netlify.app/.git/config`, `/.git/HEAD` and `/.git/logs/HEAD` all returned **200**, meaning anyone could walk the reflog, read commit messages and my personal email address, and reconstruct every past version of the site — including the dead 52 MB video blob still sitting in history.

After: added `build-deploy.sh`, which copies only the four genuinely public files into a separate `ys-home-dist/` folder. That folder is what gets dragged to Netlify now, so `.git` cannot be uploaded by accident. Verified the output contains zero git files before deploying.

(For contrast: the portfolio on GitHub Pages was never exposed this way — `/.git/config` there correctly returns 404, because Pages serves from a built branch rather than the raw working directory.)

Checked and already fine

- **Every link works.** All internal links across the four portfolio pages, plus GitHub, LinkedIn, Calendly, the CV, the build log and the portfolio cross-link — all resolve. (LinkedIn returns 999 to automated requests, which is its anti-bot response, not a broken link; confirmed manually in a browser.)
- **Case-study grid** already used `repeat(auto-fit, minmax(240px, 1fr))`, so it collapses to a single column on mobile without changes.
- **HTML/CSS page weight** is tiny — the largest page is 4 KB and the shared stylesheet 2.6 KB.
- **Motion is opt-out.** The personal site's animations are wrapped in `@media (prefers-reduced-motion: reduce)`, so they stop for anyone who has asked their OS to reduce motion.

Still ugly

- The full-size screenshots are still the fix for mobile legibility; proper responsive `srcset` variants (a phone-sized crop served to phones) would be better than asking the reader to tap.
- The 52 MB blob still exists in the personal site's local git history even though it's no longer deployed. Harmless now that `.git` isn't published, but the repo is heavier than it needs to be.
- No automated check stops `.git` being redeployed — it relies on me remembering to run `build-deploy.sh` instead of dragging the source folder.